is the message sent to the target model to make it think about its secret (e.g. "Hint me"), and the *verbalizer prompt* is the question posed to the oracle (e.g. "What is the secret word in this text?"). For each combination of secret word, context prompt, and verbalizer prompt, the target processes the context prompt under its secret-word adapter, we collect $h_1, \ldots, h_K$ as in section 3.1, and the oracle answers the verbalizer prompt under the hook. Every mention of "the task" below refers to this game.

Each confidence method produces $\hat{a}$, the first of the 20 candidate words that appears in its output, and a confidence $c \in [0, 1]$. Calibration is measured against $\mathbf{1}[\hat{a} = a^\star]$, with $a^\star$ the secret word. A generation that contains none of the 20 words (including an empty generation) is scored as wrong; the rule is the same for every method, so it does not affect the comparison between methods. On Qwen3.6-27B 14% of generations are empty; on the other three oracles empty generations are rare.

The secret-word adapters for Qwen3-8B come from Karvonen et al. (2025) and those for Gemma-2-9B from Cywiński et al. (2025). We expand the previously available set of target models by training adapters for Qwen3.6-27B and Gemma-3-27B ourselves on the per-word conversation data of Cywiński et al. (2025) mixed one to one with ordinary chat data so that the target keeps behaving as an assistant. Section L gives the recipe.

3.3 The five confidence methods

Each method below takes one sample and returns a predicted word $\hat{a}$ and a confidence c. The five adapt confidence methods built for ordinary language models to an oracle that answers under the hook. We also designed three further readouts that use the steering machinery itself. All three are weaker on both calibration and ranking than every method below, on every oracle, so section J reports them in full and gives the mechanism behind each result.

(M1, baseline) Log-probability of the answer. Decode greedily under the steering hook. To align subword tokens to $\hat{a}$ we accumulate each generated token's decoded character span, locate $\hat{a}$ via a word-boundary regex, and select the tokens whose spans overlap the match. The confidence is the joint probability of those tokens:

$$c_{\text{lp}} = \prod_{t \in \text{tok}(\hat{a})} p(x_t \mid x_{<t}).$$

This adapts the self-evaluation family of Kadavath et al. (2022) to the answer word. An offset-free variant (joint probability of the first $|\text{tok}(\hat{a})|$ generated tokens) agrees within 0.02 AUROC (section A).

(M2) Temperature bootstrap. Ask the oracle the same question twenty times with a bit of randomness, see which answer comes up most often, and treat how often the top answer won as the confidence. Precisely: draw $k{=}20$ samples at temperature T, reduce each to the first candidate word it contains, and report the most frequent answer with confidence equal to its empirical frequency:

$$c_{\text{boot}} = \frac{1}{k} \sum_{i=1}^{k} \mathbf{1}[\hat{a}_i = \text{mode}(\hat{a}_{1:k})].$$

This is self-consistency (Wang et al., 2022) where the trace is a single answer word. We sweep $T \in \{0.3, 0.5, 0.7, 1.0, 1.3, 1.5\}$ on all oracles and extend to $\{1.75, 2.0\}$ on the two oracles whose calibration was still improving at $T{=}1.5$.

(M3) Direct numeric self-report. Two turns: turn 1 greedy-decodes the answer under the hook; turn 2 appends "On a scale of 0 to 100, how confident are you?" and greedy-decodes under the same hook. The response is parsed as an integer in $[0, 100]$ and divided by 100.

(M4) Constrained label scoring. Same question as M3, but the oracle is not allowed to generate freely. It is asked to reply with exactly one of five labels (very low, low, medium, high, very high), and we score the five labels with the model's own logits by forcing each label after the prompt, using constrained decoding. Three readouts of the label distribution are evaluated: the probability-weighted expected value, the mass on "very high", and the mass on "high" plus "very high". Prompt and readout details are in section K.

(M5) Forced choice. Score all 20 candidate words directly under the steering hook: force each word as the answer with constrained decoding, sum its token log-probabilities, and normalize over the 20 candidates. The answer is the top-scoring word and the confidence is its normalized probability. This removes the extraction step and the free text decode entirely. It requires knowing the candidate set, which the other methods do not.

4 Experimental Setup

Models and oracles. We evaluate four activation oracles. For Qwen3-8B and Gemma-2-9B we use the oracle checkpoints released by Karvonen et al. (2025); the secret-word target models come from the same release for Qwen3-8B and from Cywiński et al. (2025) for Gemma-2-9B. For Gemma-3-27B we use a community oracle trained with the same recipe, paired with secret-word target models that we retrain. For Qwen3.6-27B no oracle existed prior to this work: section M describes the fixes needed to train it.

Layer readout. We inject activations at layer 1 of the oracle and read from a single layer of the target: 50% depth on Qwen3-8B (layer 18 of 36, matching Karvonen et al., 2025), 75% on Qwen3.6-27B (layer 48 of 64), 25% on Gemma-2-9B (layer 10 of 42), and 75% on Gemma-3-27B (layer 46 of 62). Each oracle is trained on activations from $\{25\%, 50\%, 75\%\}$ depth, but at inference only some of those reads carry a usable secret-word signal; on Qwen3.6-27B the 50% read yields 6% task accuracy versus 41% at 75%. Section H reports the full per-layer sweep on all four bases and how the read layers were chosen.

Prompts and protocol. We use the 100 context prompts and three verbalizer prompts of Karvonen et al. (2025). Each triple of secret word, context prompt, and verbalizer prompt makes one sample, for $20 \times 100 \times 3 = 6{,}000$ samples per method per oracle. Each method runs at every setting of its temperature grid, giving 19 to 21 method rows per oracle (21 where the extended bootstrap temperatures were run); the full grid is in section A. Hardware, software, and runtime are reported in section N. The repository linked in the abstract contains the benchmark code, the patched trainer, and the 40 retrained secret-word target models, 20 for each of the two bases we retrained.

Evaluation metrics. Accuracy is exact match between $\hat{a}$ and $a^\star$. Expected calibration error (Naeini et al., 2015, ECE) with 10 equal-width bins measures calibration: whether the model is right 80% of the time when it reports 80% confidence. Brier score (Brier, 1950) and negative log-likelihood (NLL) are proper scoring rules; they jointly penalize miscalibration and poor discrimination and cannot be gamed by a constant predictor. AUROC of confidence as a predictor of correctness measures ranking: do high-confidence outputs outrank low-confidence ones? A method can game ECE alone by emitting a constant near the overall accuracy, so we report both axes.

5 Results

5.1 Calibration results

Table 1 reports accuracy, ECE, and AUROC for the five methods on all four oracles, with one bootstrap standard error on ECE and AUROC; fig. 2 in the appendix plots the calibration-versus-ranking plane. The full grid with all five metrics is in section A, bootstrap 95% confidence intervals in section B, and reliability diagrams in section E.

Forced choice has the best AUROC on every oracle. On calibration, the tuned bootstrap is the only free-text method inside ECE 0.12 on all four oracles; which method has the single lowest ECE varies from oracle to oracle, but only among three contenders. Log-probability beats the bootstrap on Gemma-2-9B (0.037 vs. 0.040, overlapping intervals, section B) while separating correct from wrong answers weakly (AUROC 0.666), and the constrained-label readout has the lowest ECE on the two 27B oracles (0.069, tied with bootstrap T=2.0 at 0.077 on Qwen3.6-27B; 0.088 on Gemma-3-27B) while sitting at ECE 0.425 on Qwen3-8B. Direct numeric self-report has the weakest calibration and the weakest separation on every oracle. Given that the four oracles span two model families and a 3.3× size range, we read this consistency as a property of the methods themselves, though all four oracles share one training recipe.

5.2 When the candidate set is known

When the auditor knows the candidate answers, the free-text pipeline (decode, extract a word, attach a confidence) can be replaced by scoring the candidates under the injected activation. On this task the effect is large. Forced choice raises accuracy from 0.414 to 0.646 on Qwen3-8B, from 0.234 to 0.533 on Qwen3.6-27B, from 0.217 to 0.590 on Gemma-2-9B, and from 0.433 to 0.550 on Gemma-3-27B. Its confidence also separates correct from wrong answers best of all methods (AUROC 0.940, 0.936, 0.922, 0.959), with ECE 0.078 to 0.131 before any rescaling. The oracle evidently knows more than the single answer it writes out: its probabilities over the 20 candidates carry information its generated text does not. Jakkli et al. (2026) conjectured this from a 15-sample multiple-choice probe; we

Method	Qwen3-8B			Qwen3.6-27B			Gemma-2-9B			Gemma-3-27B		
	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
Forced choice (M8)	.646	.078 ±.004	**.940 ±.003**	.533	.080 ±.004	**.936 ±.003**	.590	.090 ±.004	**.922 ±.003**	.550	.131 ±.004	**.959 ±.003**
Log-prob (offset)	.414	.255 ±.005	.840 ±.005	.234	.131 ±.005	.835 ±.006	.217	**.037 ±.004**	.666 ±.008	.433	.151 ±.005	.905 ±.004
Bootstrap T=1.0	.402	**.057 ±.005**	.829 ±.005	.225	.147 ±.004	.851 ±.007	.219	.177 ±.005	.724 ±.008	.429	.134 ±.005	.887 ±.004
Bootstrap T=1.5	.367	.083 ±.005	.824 ±.006	.206	.103 ±.004	.812 ±.007	.214	.054 ±.004	.748 ±.007	.416	.131 ±.005	.864 ±.005
Constr. label P(v. high)	.414	.425 ±.006	.812 ±.005	.232	**.069 ±.005**	.778 ±.006	.217	.181 ±.005	.635 ±.009	.433	**.088 ±.005**	.813 ±.005
Direct (numeric)	.414	.582 ±.006	.516 ±.003	.234	.753 ±.006	.404 ±.007	.217	.783 ±.005	.500 ±.000	.433	.562 ±.007	.531 ±.003

Table 1: Scorecard on all four oracles, n=6,000 samples per row per oracle. ECE and AUROC carry ± one bootstrap standard error (1000 resamples). Lower ECE/higher accuracy and AUROC are better; best ECE and AUROC in **bold**. The three readouts built for the steered setting are in table 13; the full method grid is in section A.

confirm it at 6,000 samples per oracle.

5.3 Free text without labels: bootstrap

When the output must remain free text and no labeled examples exist, the bootstrap agreement rate is the most dependable confidence on every oracle. Table 2 sweeps the sampling temperature. Here, temperature matters, as the agreement rate estimates how likely a fresh sample is to repeat the most common answer, and the confidence is calibrated when that probability matches how often the oracle is actually right. Raising the temperature spreads the samples out and lowers the agreement, so on an oracle that is harder to read, the randomness has to be turned up higher: the best ECE sits at T=1.0 on Qwen3-8B (accuracy 0.40), at T=1.3 on Gemma-3-27B (accuracy 0.42), at T=1.75 on Gemma-2-9B (accuracy 0.20), and is still falling at T=2.0 on Qwen3.6-27B (accuracy 0.17). Gemma-3-27B optimizing at T=1.3 despite the highest accuracy shows there is no fixed rule from accuracy to temperature, so in practice choose T so that the mean agreement rate on a held-out slice of secret words matches the accuracy on that slice. On all four oracles, this choice lands on the best ECE of the grid. Recomputing the confidence from the first k of the 20 saved draws (section D) shows calibration and ranking are mostly in place by k=10 on all four oracles; at k=3 the mode is too noisy an estimator (ECE 0.22 to 0.31).

5.4 Free text with labels: log-prob

Once labeled examples are available, the picture changes. Fitting a post-hoc calibrator (temperature scaling, Platt, isotonic, or beta; section C) on a random half of the data brings nearly every method to ECE 0.01 to 0.03 on every oracle, including the ones whose raw confidence points the wrong way. This is expected: the calibrator is itself fit on labels and can map even a weak score onto roughly correct probabilities. What a rescale cannot change is the ordering of the scores, so once labels exist AUROC is what distinguishes methods, and there log-probability (0.84, 0.84, 0.67, 0.91) is close to the bootstrap at one generation instead of twenty. With no labels, the bootstrap agreement rate is the only confidence that is calibrated as-is; with labels, rescaled log-probability gives comparable calibration at 1/20th of the cost.

Two caveats from the word-disjoint split, which fits the calibrator on 10 secret words and tests on the other 10 (section C). Word identity carries systematic difficulty, so the word-disjoint ECE exceeds the random-split ECE on nearly every row. And a calibrator can make an already calibrated method worse: on Qwen3-8B, bootstrap at T=1.0 moves from ECE 0.061 to 0.094 under the word-disjoint isotonic fit, because the calibrator overfits word-level frequency structure.

5.5 Self-report and constrained labels

Direct numeric self-report assigns nearly the same confidence to correct and wrong answers on every oracle. The mean confidence gap between correct and wrong is +0.003 on Qwen3-8B, −0.013 on Qwen3.6-27B, 0.000 on Gemma-2-9B (the model answers "100" on essentially every sample), and +0.006 on Gemma-3-27B. AUROC is 0.516, 0.404, 0.500, and 0.531: no usable signal anywhere, and below chance on Qwen3.6-27B, whose oracle is on average more confident in its wrong answers. We attribute this to habit overriding evidence, a pattern already documented on unmodified models, whose verbalized confidence clusters at high values regardless of correctness (Mielke et al., 2022; Xiong et al., 2024): in chat data, "how confident are you?" is almost always followed by a high number, and the oracle reproduces that pattern regardless of what the injected activation supports. This replicates, on steered oracles, the gap reported by Yuan et al. (2026) and Miao and Ungar (2026) on standard models: the activations carry an uncer-

	Qwen3-8B		Qwen3.6-27B		Gemma-2-9B		Gemma-3-27B	
T	Acc	ECE	Acc	ECE	Acc	ECE	Acc	ECE
0.3	.419	.334	.233	.479	.218	.613	.434	.453
0.5	.424	.193	.231	.327	.218	.480	.435	.362
0.7	.414	.097	.230	.218	.222	.338	.435	.260
1.0	.402	**.057**	.225	.147	.219	.177	.429	.134
1.3	.387	.076	.209	.125	.219	.079	.422	**.117**
1.5	.367	.083	.206	.103	.214	.054	.416	.131
1.75	–	–	.188	.097	.202	**.040**	–	–
2.0	–	–	.172	**.077**	.204	.050	–	–

Table 2: Bootstrap temperature sweep, k=20 samples per item, best ECE per oracle in **bold**. The extended grid (T>1.5) was run on the two oracles whose ECE was still falling at T=1.5: the optimum is interior on Gemma-2-9B (T=1.75) and still falling at T=2.0 on Qwen3.6-27B.

tainty signal that the model's spoken confidence does not.

The constrained-label readout (M4) skips free generation entirely and recovers a usable signal on all four oracles: best-readout AUROC 0.819 on Qwen3-8B, 0.819 on Qwen3.6-27B, 0.680 on Gemma-2-9B, and 0.821 on Gemma-3-27B, against 0.40 to 0.53 for the free-form question. The mass on "very high" is also the best-calibrated single number on both 27B oracles (ECE 0.069 and 0.088), though it reaches ECE 0.425 on Qwen3-8B, so it is a strong readout on some oracles rather than a dependable default. An earlier version of this work reported a 30-sample pilot of this method at AUROC 0.957; the full evaluation settles near 0.82, so the pilot was optimistic, and we report the full-scale numbers as the result.

5.6 Word difficulty is uneven

Per-word accuracy spans an order of magnitude on every oracle, for example 0.12 to 0.83 on Qwen3-8B (section F), and word difficulty is only partly shared across oracles, which is why the word-disjoint calibration split of section 5.4 is the harder one. Wrong answers fall into two modes. Generic tokens ("secret", "password", empty output) carry low token probability and are easy to down-rank; committed topical neighbors ("flame" → "fire", "song" → "music"), common on Qwen3-8B, are the real issue because the oracle is confidently wrong in a plausible way. This is part of why AUROC on Qwen3-8B is lower than on Qwen3.6-27B despite the higher task accuracy. A controlled variant restricting the vocabulary to $N \in \{2, 5, 10, 20\}$ words is in section G.

6 Conclusion

We measured five ways of producing a confidence score on four activation oracles spanning two model families and a 3.3× size range, at 6,000 samples per method and oracle, for 120,000 scored answers in the main comparison and more in the temperature sweeps and the appendix readouts. The five methods rank the same way on all four oracles.

We release two artifacts with the paper. The first is a testbed: the four oracles, twenty model organisms for each, the sample protocol, and a scorer that takes a list of answer and confidence pairs and returns accuracy, ECE, Brier, negative log-likelihood, and AUROC. New confidence methods were created or evaluated specifically for activation oracles, both from the literature on confidence estimation methods and from demo attempts. The second artifact is the Qwen3.6-27B oracle, with its reproducible recipe, to our knowledge the first activation oracle for a hybrid attention architecture, which we release with the trainer changes it required and the forty secret-word target models we retrained.

Recommendations.

- If the hypothesis space can be enumerated, score the candidates directly under the injection (M5); the normalized probability is the best-ranking confidence we measured.
- For free-text answers with no labeled data, use bootstrap mode frequency. Tune T so that mean mode frequency on a held-out word slice matches its empirical accuracy; k=10 samples capture most of the value of k=20.
- Once labeled data exists, fit a rescale (isotonic or beta) on the answer-word log-probability when the twenty-decode budget of the bootstrap is too expensive.
- Do not read free-form numeric self-report as a probability; prefer constrained label scoring or a confidence head.